MIL-W-50258A (MU)
15 November 1971
SUPERSEDING
MIL-W-50258 (MU)
28 February 1969

MILITARY SPECIFICATION

WARHEAD SECTION, 762MM ROCKET,

PRACTICE: M38

METAL PARTS FOR

1. SCOPE

1.1 This specification covers the metal parts for one type of warhead section designated as Warhead Section, 762MM Rocket, Practice: M38.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATION

FEDERAL

QQ-P-416	-	Plating, Cadmium (Electrodeposited)
QQ-Z-325	-	Zinc Plating (Electrodeposited)

MILITARY

MIL-A-2550	-	Ammunition and Special Weapons, General Specification for

FSC: 1340

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MIL-C-5541	-	Chemical Treatment of Aluminum and Aluminum Alloys
MIL-I-6866	-	Inspection, Penetrant, Method of
MIL-I-45607	-	Inspection Equipment, Acquisition Maintenance and Disposition of
MIL-P-116	-	Preservation, Method of
MIL-T-5021	-	Test, Aircraft Welding Operators, Certification of
MIL-W-8604	-	Welding of Aluminum Alloys Process for

STANDARD

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109	-	Quality Assurance Terms and Definitions
MIL-STD-453	-	Inspection, Radiographic
MIL-STD-1168	-	Lot Numbering of Ammunition
MIL-STD-1235	-	Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

US ARMY MUNITIONS COMMAND

8824941	-	Shipping & Storage Container, Warhead Section M473A1 for 762MM Rocket, Practice M38 Packing and Marking

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8833470 - Metal Parts Assembly

PUBLICATIONS

ARMY

ET-8833470 - Equipment Tabulation

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

2.2 Other publications.-The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

CODE OF FEDERAL REGULATIONS

Title 49 - Transportation, Parts 0-190

(The Code of Federal Regulations is available from the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402. Orders should specify 49 CFR 0-190 (latest revison).)

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM E
8-69 (Part 31) - Standard Methods of Tension Testing of Metallic Materials

(Application for copies should be addressed to: American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103)

3. REQUIREMENTS

3.1 Material.-Materials shall be in accordance with the applicable drawings and specifications.

3.2 Parts.-The parts and assemblies shall comply with all requirements specified on Drawing (Dwg) 8833470, all associated drawings, and with all requirements specified in applicable specifications.

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3.3 Welding.-The weld of each warhead section shall be such as to produce satisfactory joined parts. All welds shall be reasonably smooth and free of craters. They shall exhibit characteristics of good fusion, penetration and soundness of deposit representative of good welding practice. The process shall be such as to prevent warpage beyond dimensional requirements. The satisfactoriness of the weld shall be based upon compliance with 3.3.1, 3.3.2, 3.3.3, 3.3.4, 3.3.5 and 3.4.2.

3.3.1 Electrodes.-Electrodes shall be of a coated type, except as specified in 3.3.2. The coating shall be uniform and without bare spots, except those used to introduce current to the electrode. All types of electrodes to be used shall be approved by the Government.

3.3.2 Process.-Gas shielded arc-welding or other automatic arc-welding in which bare or lightly coated wire is used with the arc bathed in a reducing gas or other atmosphere excluding media, may be used.

3.3.3 Qualification.

3.3.3.1 Welding operators.-Welding operators shall be qualified in accordance with MIL-T-5021.

3.3.3.2 Welding equipment.-Welding equipment shall be qualified in accordance with MIL-W-8604.

3.3.3.3 Welding Procedure.-All welding shall be accomplished in conformance with the process used to qualify the first article sample.

3.3.4 Acceptable weld imperfections permitted in warhead section.

3.3.4.1 Sag.-Sag of the material in the area adjacent to the weld shall not be more than .010 inch below the surface of the joined part except ballast weldment.

3.3.4.2 Porosity.-Porosity or other discontinuities shall not extend as surface cracks in excess of those defined in 3.3.5 as determined by examination in accordance with 4.3.3.1, except that microporosity as evidence by partial penetration into skins by penetrant, shall be acceptable.

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3.3.4.3 Blow-outs.-Blow-outs may be repaired by hand welding operations, provided they are no greater than one quarter (1/4) inch in width and one (1) inch in length and not more than three (3) such areas may be repaired in any one skin.

3.3.5 Surface cracks Aluminum.-Surface cracks shall not be in excess of the following:

a. There shall be no more than one (1) surface crack 1/2 inch long in each one (1) inch of weld.

b. There shall be no more than a total of two (2) inches of surface cracks in each weld.

3.4 Mechanical properties.

3.4.1 Component parts.-The aluminum used in the fabrication of the component parts shall comply with the minimum (min) properties specified on the applicable drawing and the heat treat condition required, when tested as specified in 4.2.2.1 or 4.3.3.2, as applicable.

3.4.2 Welds (aluminum).-The mechanical properties of the welds shall not be less than 30,000 pounds per square inch (psi) tensile strength when tested as specified in 4.2.2.2. Unless otherwise directed by the Contracting Officer, this test shall be performed on the first article sample only.

3.5 Radiographic examination.-Unless otherwise specified by the procuring agency, radiographic examination shall be made in accordance with MIL-STD-453.

3.6 Functioning of ogive.-The ogive, shall function satisfactorily when tested as specified in 4.4.1.

3.7 Functioning of pyrocore booster assembly.-The pyrocore booster assembly shall function satisfactorily as evidenced by complete detonation of the primer when tested as specified in 4.4.2.

3.8 Functioning of Adapter Assembly, Fuze Holder.-The adapter assembly, fuze holder shall function satisfactorily when tested as specified in 4.4.10.

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3.9 Heat-sealed Seam Test.-The heat-sealed seam may evidence partial separation within the first two (2) minutes of test. No additional separation permitted in the heat-sealed area during the final three (3) minutes of the test. The test shall be performed as specified in 4.4.11.

3.10 Workmanship.-All parts and assemblies shall be fabricated, finished, and assembled in a thorough, workmanlike manner. They shall be free of burrs, chips, sharp edges, cracks, unblended radii, surface defects, dirt, grease, rust, corrosion products and other foreign matter. The cleaning method used shall not be injurious to any part, nor shall the parts be contaminated by the cleaning agents. Surface coatings shall be continuous except for a few small scratches not exposing base material. All required markings shall be neat and sharply defined.

3.11 Submission of results of contractor-conducted examinations and tests.-The contractor shall forward copies of the results of any of his examinations or tests to the designated address as required by the contract. (See 6.4)

3.12 First article testing.-This specification makes provisions for first article testing. Submission of first article quantity by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 to define terms used herein. The provisions of MIL-A-2550 shall apply.

4.1.1 Submission of product.-At the time the completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the

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following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all the quality assurance provisions specified in this specification.
b. Drawing and specification number and date, together with identification and date of changes thereto.
c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.
d. Number of items in the lot.
e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 First article inspection

4.2.1 Submission.-The contractor shall submit a first article quantity as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.2.2. (See 6.1) The first article sample shall consist of two (2) complete assemblies and three (3) complete sets of parts (i.e. three (3) each of every component and every sub-assembly). All parts and completed assemblies submitted shall have been produced by the contractor (or furnished by a supplier) using the same production processes, procedures and equipment as will be used in fulfilling the contract. All parts and materials, including packaging and packing, shall be obtained from the same sources of supply as will be used in regular production. The sample shall be accompanied by certificates of conformance. A first article quantity, or portion thereof, as directed by the Contracting Officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days, or whenever a change occurs in manufacturing process, material used, drawing, specification or sources of supply as to significantly affect product uniformity as determined by the Government. Prior to submission,

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the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample. A sample containing known defects will not be submitted unless specifically authorized by the Contracting Officer.

4.2.2 Inspections to be performed.-Assemblies and components will be subjected by the Government to any or all the examinations or tests specified in this specification and any or all requirements of the applicable drawings. In addition, any or all of the following examinations and tests will be performed:

4.2.2.1 Mechanical properties of component parts (Destructive test).-Sixteen (16) specimens complying with ASTM E8-69 (Part 31), shall be taken from one (1) warhead for tensile and yield strength tests. The 16 specimens shall be taken as follows: Four (4) shall be taken from each skin; of each set of four, two (2) shall be taken parallel to the longitudinal axis and two (2) normal to the axis of the skin. Test shall be in accordance with ASTM E8-69 (Part 31) using the offset method.

4.2.2.2 Tensile properties of skin welds (Destructive tests).-Four (4) samples complying with the type specified in ASTM E 8-69 (Part 31) shall be taken transverse to the welded joint from each skin. Two (2) samples shall be taken parallel to longitudinal axis of the welded joint from each skin. Each weld shall be tested for conformance to the requirements of 3.4.2. Using method specified in ASTM E8-69 (Part 31).

4.2.3 Rejection.-If any assembly or component fails to comply with any of the applicable requirements, the first article quantity shall be rejected. The Government reserves the right to terminate its inspection upon any failure of an assembly or component in the sample to comply with any of the stated requirements. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit a new first article quantity or portion thereof. Until a first article quantity is accepted, the contractor is in no way authorized by the Government to resume regular production unless otherwise directed by the Contracting Officer.

4.3 Inspection provisions.

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4.3.1 Lot formation.-The term "inspection lot" as used in this specification is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to process, specification, or drawing not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168.

4.3.2 Examination.-The examinations listed in the following subparagraphs shall be performed on inspection lots (See 4.3.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for critical defects (and major defects, when so specified) shall be 100 percent. Sampling plans for major and minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's designated herein to the groups of major and minor defects listed, the contractor may elect to apply AQL's to individual defects using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. When this method is elected, it shall be applied to all of the major and minor defects listed in the applicable subparagraph of this section, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

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4.3.2.1 Skin, base (See Dwg 8833459 covering a detail of Dwg 8833467).

Categories	Defects	Method of Inspection	Code No. (See 6.2)
Critical: None defined			
Major: None defined			
Minor	AQL 1.50 Percent		
201.	Length	Gage	01001
202.	Thickness	Gage	01002
203.	Evidence of poor workmanship (See 3.10)	Visual	01003

4.3.2.2 Skin, forward (See Dwg 8833463 covering a detail of Dwg 8833466).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major: None defined			
Minor	AQL 1.50 Percent		
201.	Length	Gage	02001
202.	Contour	Gage	02002
203.	Thickness	Gage	02003
204.	Evidence of poor workmanship (See 3.10)	Visual	02004

4.3.2.3 Skin, rear (See Dwg 8833455 covering a detail of Dwg 8833466).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major: None defined			
Minor	AQL 1.50 Percent		
201.	Length	Gage	03001
202.	Contour	Gage	03002

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203.	Thickness	Gage	03003
204.	Evidence of poor workmanship (See 3.10)	Visual	03004

4.3.2.4 Skin, ogive (See Dwg 8833465 covering a detail of Dwg 8833448).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	None defined		
Minor	AQL 1.50 Percent		
201.	Length	Gage	04001
202.	Thickness	Gage	04002
203.	Depth, min	Gage	04003
204.	Evidence of poor workmanship (See 3.10)	Visual	04004

4.3.2.5 Frame, separation (See Dwg 8833476 covering a detail of Dwg 8833466).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	None defined		
Minor	AQL 1.00 Percent		
201.	Height	Gage	05001
202.	Evidence of poor workmanship (See 3.10)	Visual	05002

4.3.2.6 Base assembly (See Dwg 8833467 covering a detail of Dwg 8833461).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major	AQL 1.00 Percent		
101.	Location of mounting hole	Gage	06001

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102.	Location of pin hole	Gage	06002
103.	Pitch diameter of mounting hole thread	Gage	06003
104.	Minor diameter of mounting hole thread	Gage	06004
105.	Depth of pin hole, min	Gage	06005
106.	Diameter of pin hole, min	Gage	06006
107.	Base fitting assembly missing	Visual	06007
Minor	AQL 1.00 Percent		
201.	Marking missing or misleading	Visual	06008
202.	Evidence of poor workmanship (See 3.10)	Visual	06009

4.3.2.7 Base and ballast assembly (See Dwg 8833461 covering a detail of Dwg 8833470).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.65 Percent		
101.	Depth of charge well cavity	Gage	07001
102.	Diameter of charge well cavity	Gage	07002
103.	Lifting ring loose	Manual	07003
Minor	AQL 1.00 Percent		
201.	Screw missing	Visual	07004
202.	Evidence of poor workmanship (See 3.10)	Visual	07005

4.3.2.8 Booster lead cup (See Dwg 8833424).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major	AQL 0.65 Percent		
101.	Radius under flange	Gage	36001
102.	Flange diameter	Gage	36002
103.	Cracks	Visual	36003
Minor	AQL 0.65 Percent		
201.	Evidence of poor workmanship (See 3.10)	Visual	36004

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4.3.2.9 Booster lead cup assembly (See Dwg 8833428 covering a detail of Dwg 8833434).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major	AQL 0.65 Percent		
101.	Distance of charge incorrect	Gage	08001
102.	O.D. max	Gage	08003
103.	Distance from under flange to bottom (max)	Gage	08004
104.	Cracks	Visual	08005
Minor	AQL 0.65 Percent		
201.	Evidence of poor workmanship (See 3.10)	Visual	08002

4.3.2.10 Pyrocore Booster Assembly (See Dwg. 8833456).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major	AQL 0.40 Percent		
101.	Braided lead wire not fully extended into end primer	Visual	33001
Minor	AQL 0.65 Percent		
201.	Evidence of poor workmanship (See 3.10)	Visual	33002

4.3.2.11 Confining Plate (See Dwg. 8833452).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major	AQL 0.40 Percent		
101.	Radius (2 places)	Gage	34001
102.	Location of Radii	Gage	34002
Minor	AQL 0.65 Percent		
201.	Evidence of poor workmanship (See 3.10)	Visual	34003

4.3.2.12 Holder (See Dwg 8833447 covering a detail of Dwg 8833433).

Categories	Defects	Method of Inspection	Code No.

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Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major	AQL 0.40 Percent		
101.	Location of flash holes	Gage	09001
Minor	AQL 0.65 Percent		
201.	Evidence of poor workmanship (See 3.10)	Visual	09002

4.3.2.13 Lead holder assembly, prior to assembling disc (See Dwg 8833434 covering a detail of Dwg 8833475).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major	AQL 0.40 Percent		
101.	Booster lead cup assembly missing	Visual	10001
Minor	AQL 0.65 Percent		
201.	Evidence of poor workmanship (See 3.10)	Visual	10002

4.3.2.14 Fuze ring assembly (See Dwg 8833475 covering a detail of Dwg 8833466).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	None defined		
Minor	AQL 1.50 Percent		
201.	Pyrocore cover loose	Manual	11001
202.	Gasket missing	Visual	11002
203.	Absence of adhesive coating on primer lead	Visual	11004
204.	Evidence of poor workmanship (See 3.10)	Visual	11003

4.3.2.15 Body assembly (See Dwg 8833466 covering a detail of Dwg 8833470).

Categories	Defects	Method of Inspection	Code No.

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Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major	AQL 0.65 Percent		
101.	Lead holder assembly missing or loose	Visual-Manual	12001
102.	Index loose	Visual-Manual	12002
103.	Pyrocore booster assembly missing	Visual	12003
104.	Fuze ring insulator missing	Visual	12004
Minor	AQL 1.50 Percent		
201.	Clamp missing or loose	Visual-Manual	12005
202.	Lubricant missing from cam contacting area	Visual	12006
203.	Stake or sealing compound missing from lead holder assembly	Visual	12007
204.	Evidence of poor workmanship (See 3.10)	Visual	12008

4.3.2.16 Ogive, rocket (See Dwg 8833448 covering a detail of Dwg 8833470).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	None defined		
Minor	AQL 1.00 Percent		
201.	Gasket improperly secured	Visual	13001
202.	Evidence of poor workmanship (See 3.10)	Visual	13002

4.3.2.17 Metal parts assembly, prior to painting (See Dwg 8833470).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major	AQL 1.50 Percent		
101.	Concentricity and perpendicularity of the first datum (base end) with the mounting holes	Gage	14001
102.	Concentricity and perpendicularity of second datum with the mounting holes	Gage	14002

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103.	Concentricity and perpendicularity of third datum with the mounting holes	Gage	14003
104.	Concentricity and perpendicularity of fourth datum with the mounting holes	Gage	14004
105.	Concentricity and perpendicularity of fifth datum with the mounting holes	Gage	14005
106.	Outside diameter, max	Gage	14006
107.	Screw or rivet head above or below skin surface excessive	Gage	14007
108.	Nick, scratch, dent, cut or protrusion above or below skin surface excessive	Gage	14008
109.	Length from base ring to base skin, max	Gage	14009
110.	Gap, max	Gage	14010
111.	Mismatch between external surfaces, excessive	Gage	14011
112.	Welds above skin surface excessive	Gage	14012
113.	Primer depth incorrect	Gage	14013
114.	Ogive assembly loose	Manual	14014
115.	Index, rod, missing	Visual	14015
116.	Fuze lock-up system inoperable	Manual	14016
117.	Fuze holder adapter assembly missing	Visual	14017
118.	Tape missing	Visual	14021
119.	Pyrocore Broken	Visual	14022
120.	Presence of adhesive on end surface of primer	Visual	14023
Minor	AQL 1.50 Percent		
201.	Total length, max	Gage	14018
202.	Screw, missing or loose	Visual-Manual	14019
203.	Absence of adhesive on primer lead	Visual	14024
204.	Evidence of poor workmanship (See 3.10)	Visual	14020

4.3.2.18 Bag, prior to sealing (See Dwg 8833470).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		

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Major: None defined

Minor		AQL 1.50 Percent	
201.	Plug missing	Visual	15001
202.	Bolt missing	Visual	15002
203.	Washer missing	Visual	15003
204.	Evidence of poor workmanship (See 3.10)	Visual	15004

4.3.2.19 Metal parts assembly (See Dwg 8833470).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major	AQL 1.00 Percent		
101.	Weight	Balance	16001
102.	Pressure diaphram missing, improperly assembled or damaged	Visual	16002
103.	White squares missing	Visual	16003
104.	Bag torn or perforated	Visual	16004
105.	Bag improperly secured	Visual	16005
Minor	AQL 4.00 Percent		
201.	Bare spot on exterior surface, except for slight scratches	Visual	16006
202.	Marking misleading or unidentifiable	Visual	16007
203.	Closing disc perforated	Visual	16008
204.	Marking improperly located	Visual	16009
205.	Closing disc improperly secured	Visual	16010
206.	Ring assembly missing	Visual	16011
207.	Primers improperly secured	Visual	16012
208.	Evidence of poor workmanship (See 3.10)	Visual	16013

4.3.2.20 Shipping and storage container, prior to assembly of cover (See Dwg 8824941).-

Categories	Defects	Method of Inspection	Code No.

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Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major	AQL 1.00 Percent		
101.	Nose strap missing	Visual	17001
102.	Bolts missing	Visual	17002
103.	Container for fuze missing or damaged	Visual	17004
104.	Desiccant bags missing or insecure	Visual	17005
105.	Barrier bag assembly torn or perforated	Visual	17006
106.	Humidity indicator missing	Visual	17007
Minor	AQL 1.50 Percent		
201.	Pad loose	Visual	17008
202.	Bracket support loose	Visual	17009
203.	Nose strap loose	Visual	17010
204.	Evidence of poor workmanship (See 3.10)	Visual	17003

4.3.2.21 Shipping and storage container (See Dwg 8824941).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major	AQL 0.40 Percent		
101.	Quicklock missing	Visual	18001
102.	Hardware missing or damaged	Visual	18002
Minor	AQL 2.50 Percent		
201.	Marking misleading or unidentifiable	Visual	18003
202.	Wood damaged or loose	Visual	18004
203.	Socket head screw key missing	Visual	18005
204.	Side, end or top improperly assembled	Visual	18007
205.	Evidence of poor workmanship (See 3.10)	Visual	18006

4.3.3 Testing.

4.3.3.1 Penetrant examination of skin welds (See 3.3.4.2) Major Defect, Code No. 19001.-The weld of each skin shall be examined as specified in 4.4.5.

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4.3.3.2 Mechanical properties of component parts, Major Defect, Code No. 20001.-After acceptance of the first article sample, the mechanical properties of component parts shall be determined on test specimens of the same material and heat treatment as that used in the warhead section. Tests shall be made in accordance with ASTM E8-69 (Part 31).

4.3.3.3 Functioning of ogive (See 3.6) Major Defect, Code No. 21001.-The ogive, assembled to each warhead section shall be tested for proper functioning as specified in 4.4.1. Any warhead section failing to meet this requirement shall be classified defective and removed from the lot.

4.3.3.4 Lateral center of gravity (See Dwg. 8833470) Major defect, Code No. 22001.-Sampling for inspection of lateral center of gravity shall be conducted in accordance with MIL-STD-1235. The continuous sampling plan to be applied shall be CSP-M, using an AQL of 5.0% and a frequency (f) of 1/3. The production process shall meet the requirements of paragraph 8.6.2 of MIL-STD-1235. 100% inspection shall be conducted until ten (10) consecutive items have been found acceptable. At this point inspection shall be reduced to 1/3 of the items produced consecutively. When, during inspection at this level (1/3), ten (10) items have been inspected without finding a defective item, the level of inspection shall be reduced to 1/9. This level shall be maintained as long as no defective items are found. If a defective item is found during the inspection above, the inspection plan shall be modified as required by Figure 4, Page 21 of MIL-STD-1235 (17 July 1962). Testing shall be as specified in 4.4.6.

4.3.3.5 Longitudinal center of gravity, (See Dwg 8833470), Major Defect, Code No. 23001.-Sampling inspection of longitudinal center of gravity shall be conducted using the sampling plan described in 4.3.3.4. Unless otherwise determined by the Government, the same sample items may be used for this test as were used under 4.3.3.4. The test shall be performed as specified in 4.4.7.

NOTE: The requirements of Paragraph 8.6.5 of MIL-STD-1235 shall not apply to the inspection required by 4.3.3.4 and 4.3.3.5. Instead, 100% inspection shall be resumed immediately following a lapse in production of ninety (90) days, or whenever a change occurs in manufacturing process, material used, drawing, specification or source of supply as to significantly affect product uniformity as determined by the Government.

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4.3.3.6 Radiographic examination of pyrocore booster assembly (See Dwg 8833456) Major Defect, Code No. 24001.-Each pyrocore booster assembly shall be examined to determine compliance to the applicable drawing requirement. Any assembly which is found defective shall be removed from the lot. This examination shall be performed as specified in 4.4.8.

4.3.3.7 Functioning of pyrocore booster assembly (See 3.7 and Dwg 8833456) Major Defect, Code No. 25001.-After a lot of pyrocore booster assemblies have been examined in accordance with 4.3.3.6, two (2) of the assemblies shall be selected from the lot and tested as specified in 4.4.2. If either fails to function as required, the lot shall be considered defective and shall not be used in assembly.

4.3.3.8 Graphite content of tetryl (See Dwg 8833428) Major Defect, Code No. 26001.-At the time and place of loading the booster lead cups, a sample of approximately 10 grams of tetryl shall be selected from each lot and subjected to this test. If any sample fails to comply with the applicable requirements, the lot shall be rejected. The test shall be conducted as specified in 4.4.3.

4.3.3.9 Specific gravity of booster lead cup assembly charge (See Dwg 8833428) Major Defect, Code No. 27001.-A sample of three (3) loaded booster lead cup assemblies shall be selected from each four (4) hours production and tested as specified in 4.4.4. If any sample fails to conform to the drawing requirement, the four (4) hour's production shall be rejected.

4.3.3.10 Salt spray, Major Defect, Code No. 28001.-Sample specimens as specified in QQ-A-325, QQ-P-416 or MIL-C-5541, as applicable, shall be submitted for each plated part and subjected to this test; this is a destructive test, parts so tested will be scrapped.

4.3.3.11 Functioning of Adapter Assembly, Fuze Holder (See 8831260 and 3.8) Major Defect, (100% Inspection) Code No. 31001.-Each adapter assembly, fuze holder shall be tested for proper functioning as specified in 4.4.10. Any adapter assembly failing to meet specified requirements shall be classified defective and removed from the lot.

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4.3.3.12 Heat-sealed Seam Test (See 3.4 and Dwg. 8824941) Major defect, Code No. 32001.-Sampling for this test shall be one coupon approximately 6" to 12" of barrier bag material, heat-sealed, at the beginning of each lot and one coupon, heat-sealed, at the end of each lot. If the heat-sealed sample at the beginning of the lot fails to meet the heat-sealed seam test requirement, production shall not begin until a satisfactory heat seal has been accomplished. If the heat-seal seam test fails on sample heat-sealed at the end of the lot, the entire lot shall be rejected and all items in that lot shall be re-heat-sealed including sample coupon. If second (2nd) heat-sealed seam test fails, entire lot shall be rejected. The test shall be performed as specified in 4.4.11.

4.3.3.13 Booster Pyrocore Assembly missing or incorrectly installed (See Dwg 8833470) Major defect (100% Inspection) Code No. 35001.-Prior to the assembly of the diaphragm each Warhead Section shall be visually inspected for presence of the Booster Pyrocre Assembly and for the installation of each lead of the divided strands into different spotting charge wells.

4.3.4 Inspection equipment.-Equipment Tabulation (ET) number 8833470 and Equipment Lists (EL's) referenced theron identify the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in 6.3. Supply and maintenance of the equipment listed on the EL's shall be in accordance with MIL-I-45607.

4.3.4.1 Government rights to documentation.-Inspection equipment drawings and lists provided and revised in accordance with the requirements of this specification may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey, and whenever inspection equipment drawings are needed.

4.4 Test methods and procedures.

4.4.1 Functioning of ogive, rocket.-The ogive shall be mounted on the fuze ring assembly and the index rod shall be fastened in place. The latch buttons shall be depressed and

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the hooks shall be engaged in the clevises. The latches shall then be closed. At this point, the rocket, ogive, shall be firmly held in place by the latches and shall show no obvious tilt or out-of-round condition when viewed from any angle. The push button shall then be depressed, using thumb pressure. The latch levers shall be fully opened. The ogive, rocket shall be separated from the fuze ring assembly by pulling forward and turning in a clockwise direction. This is a non-destructive test. Parts so tested shall be returned to the lot.

4.4.2 Functioning of pyrocore booster assembly.-Using the equipment specified in the Equipment List, tape a Number 6 RDX electric blasting cap in a butt position to the center section of the pyrocore booster assembly. Supply sufficient voltage to detonate the blasting cap. This is a destructive test. Parts so tested shall not be returned to the lot.

4.4.3 Graphite content of tetryl.-An accurately weighed portion of five (5) grams of tetryl shall be transferred to a tared medium porosity sintered glass crucible. The sample shall be extracted with four - 10ML portions of acetone, maintaining each portion in contact with the residue for approximately five (5) minutes before applying suction. Any lumps which may persist shall be broken up and washed down the sides of the crucible with each 10ml portion of acetone to insure complete removal of the tetryl. The residue shall be aspirated until the ordor of acetone is no longer detectable; the crucible and contents shall be placed in a drying oven at 100° to 105° Centrigrade for 1/2 hour, then cooled in a desiccator and weighed. The above extracting, drying and weighing shall be repeated with additional 10ml portions of acetone until the loss in weight on subsequent extraction is less than 0.0005 gm. The percentage of graphite in the sample shall be calculated from the weight of the residue in the crucible.

4.4.4 Specific gravity of the booster lead cup assembly charge.-The specific gravity of the charge of the booster lead cup assembly shall be determined as follows:

a. Weight the empty cup in grams and record as W_e.
b. Fill the cup with water at 75°F $\pm$ 2°F, weigh and record the weight in grams as W_w.
c. Empty the cup, dry it and load with tetryl, using the process and material in use in production.
d. Weigh the loaded cup in grams and record the weight as W_1.

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e. Calculate the specific gravity using the formula.

$$\text{Specific gravity} = \frac{W_1 - W_e}{W_w - W_e}$$

4.4.5 Penetrant examination of skin welds.-The weld of each skin shall be examined as specified in MIL-I-6866. This is a non-destructive test. Parts so tested shall be returned to the lot.

4.4.6 Lateral Center of gravity.-The metal parts assembly shall be mounted in a fixture listed in the Equipment List and tested for conformance to drawing requirements. This is a non-destructive test. Parts so tested shall be returned to the lot.

4.4.7 Longitudinal Center of gravity.-The metal parts assembly shall be mounted in a fixture listed on the Equipment List and tested for conformance to drawing requirements. This is a non-destructive test. Parts so tested shall be returned to the lot.

4.4.8 Radiographic examination of pyrocore booster assembly.-The pyrocore booster assembly shall be examined using x-Ray equipment and procedures specified in MIL-STD-453. Observation shall be made to determine compliance with requirements specified in applicable drawing. This is a non-destructive test. Parts so tested shall be returned to the lot.

4.4.9 Salt spray.-The sample specimens shall be tested as specified in QQ-Z-325, QQ-P-416 or MIL-C-5541, as applicable. This is a destructive test. Parts so tested shall be discarded.

4.4.10 Functioning of Adapter Assembly, Fuze Holder.-Open adapter assembly fuze holder by placing fuze clamp lever in the open position. Seat gage, listed on the equipment list, into the adapter fuze holder and lock in place, using thumb pressure. Max. and min. gages shall be held firmly and the adapter assembly fuze holder shall exhibit no sign of deformation. This is a non-destructive test. Parts so tested shall be returned to the lot.

4.4.11 Heat-sealed Seam Test.-The heat-sealed seam test shall be performed as specified in MIL-P-116, Section 4.4.4. Equipment shall be as specified on the Equipment List.

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5. PREPARATION FOR DELIVERY

5.1 Packing.

5.1.1 Level A.-The metal parts assembly shall be packed in accordance with drawing 8824941.

5.2 Marking.-Marking shall be in accordance with Dwg 8824941 and Code of Federal Regulations, Title 49, Parts 171-179.

5.3 Shipping.-When shipments from more than one lot are shipped as a carload, each lot shall be kept separate, and division between lots clearly indicated to prevent mixing of the lots in transit.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.
b. Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.
c. Provisions for submission or first article samples.

6.2 Inspection code numbers.-The five-digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government. These code numbers are also used to correlate the characteristics cited on Equipment Lists with the inspections listed in this specification. These code numbers shall be listed on drawings of equipment designs submitted by the contractor (see 6.3.5).

6.3 Inspection equipment.-The contractor shall design inspection equipment as required by the referenced Equipment Lists (EL) in accordance with the instructions of paragraphs 6.3.1 through 6.3.5.

6.3.1 Equipment Lists.-Equipment lists indicate the availability of inspection equipment designs by showing in the "Part Number" column of the list of inspection equipment (Form SMUPA 1010) the numbers of drawings or Federal Stock numbers of existing equipment designs, or codes as indicated in paragraph 6.3.2. Design action required of the contractor is described in paragraphs 6.3.3

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and 6.3.4. The contractor will be required to prepare detailed drawings in accordance with 6.3.4 for all the equipment coded as "Contractor Design" in the "Part Number" column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.2 Equipment list codes.-The inspection equipment as defined in 6.3.3 and 6.3.4 will be designated in the EL by the following codes:

CD - Contractor design on controlled contractor format and/or commercial equipment.
MU - Army design, mandatory for use.

6.3.3 Army designs.-Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army design fall into two basic classifications: Mandatory (designated "MU") and non-mandatory (indicated by drawing or Federal Stock Number). When an equipment list references mandatory Army Designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs, with the approval of the Government, design alternate inspection equipment to facilitate his operations. Such contractor-prepared designs (see 6.3.4) or commercial equipment selections must be approved by the Government prior to fabrication or procurement of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.4 Contractor designs.-Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finishes, notes and operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, contractor de-

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signs may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above, subject to the following controls: All submitted contractor designs shall conform to Specification MIL-D-1000, Category A, Form 3. Legibility and reproducibility shall permit conventional making of clearly understandable, high-contrast reproductions. Contractors shall submit three copies of final designs as a flat set. Designs shall be submitted for approval as specified in 6.3.5.

6.3.5 Submission of contractor design.-All submitted designs shall contain a reference to the applicable 5-digit code number contained in Section 4 of this specification and the appropriate component or assembly drawing number and revision letter to which the specific design applies. Unless otherwise specified on the EL, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-QA-A-S. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract. Submission shall be as directed to the contracting officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished, normally, within one month after receipt.

6.4 Submission of results of contractor-conducted examinations and tests.-Unless otherwise specified by the Contracting Officer, the contractor shall forward requested records of examinations or tests to Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-QA-A-S.

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